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UG Program in Artificial Intelligence and Data Science

Deep Learning Case Study Report

2026-2027



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Case Study Report on Real-Time
Road Accident Detection using
Deep Learning

by

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2026-2027



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Certificate

This is to certify that the report of the Case Study entitled

Real-Time Road Accident Detection using Deep Learning

is a Bonafide work of

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ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

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Deep Learning CIAP Case Study

1. Aim

To design and study a deep learning based system that can automatically detect road accidents in real time from CCTV/traffic camera video feeds, thereby helping reduce emergency response time.

2. Title

Real-Time Road Accident Detection using Deep Learning

3. Abstract (100–150 words)

- Road accidents remain one of the leading causes of preventable fatalities worldwide, and delayed emergency response due to manual monitoring is a major contributing factor. This case study presents a deep learning based Real-Time Road Accident Detection system that automatically identifies accident events from traffic surveillance video. A Convolutional Neural Network (CNN) is used for spatial feature extraction from individual frames, and is extended with a Long Short-Term Memory (LSTM) layer to model temporal patterns across frame sequences, enabling the system to distinguish accident events from routine traffic flow. The proposed CNN-LSTM model is compared against a baseline CNN classifier and evaluated using Accuracy, Precision, Recall, F1-score and ROC-AUC. The study demonstrates how deep learning can support faster emergency alerts and contributes to SDG 11 – Sustainable Cities and Communities.

4. Introduction

- Background: Deep Learning is a subfield of Machine Learning that uses multi-layered neural networks to learn complex representations directly from raw data such as images and video frames. Traditional traffic monitoring relies heavily on manual observation by control-room personnel or delayed reporting by witnesses, which often leads to slow emergency response.
- Need for the study: On highways and low-traffic roads an accident may go unnoticed for several minutes, and every minute of delay reduces the chances of survival for victims, making automated, real-time detection from existing CCTV infrastructure a genuine safety need.
- Importance of the topic: Combining CNNs, which excel at spatial feature extraction, with sequence models such as LSTMs allows automatic detection of accidents from surveillance footage in real time, giving practical experience in designing, training and evaluating a deep learning model for a safety-critical, real-world application aligned with smart-city traffic-safety goals.

**UG Program in Artificial Intelligence and Data Science****5. Problem Statement**

- Design and develop a deep learning solution for Real-Time Road Accident Detection. The problem involves analysing the domain problem, collecting and preprocessing an appropriate accident/non-accident video dataset, selecting a suitable deep learning architecture, training and validating the model, comparing it with a baseline approach, and discussing deployment challenges, ethical considerations and real-world applicability, so that accident events can be flagged automatically and emergency services alerted with minimal delay.

6. Objectives

- 1. To understand the working of deep neural networks, specifically CNNs, for visual pattern and event recognition.
- 2. To collect and preprocess a suitable road-accident image/video dataset for training a detection model.
- 3. To design and implement a CNN-LSTM deep learning model using TensorFlow/Keras or PyTorch for accident detection.
- 4. To train, validate and compare the proposed model against a baseline CNN using standard performance metrics.
- 5. To discuss real-world deployment challenges and ethical considerations of automated accident-detection systems.

7. Literature Survey

- Summary of existing approaches: a brief survey of accident and anomaly detection techniques in traffic surveillance was carried out, covering motion/optical-flow methods, classical ML on handcrafted features, CNN-based frame classification, CNN-LSTM/3D-CNN video models, and transfer learning with pretrained backbones such as ResNet, VGG or MobileNet.

Comparison of existing methods:

Approach	Key Idea	Limitation
Motion / optical-flow based detection	Detects sudden changes in vehicle trajectory or speed	Sensitive to camera shake and lighting changes
Classical ML (SVM / Random Forest)	Uses handcrafted features such as speed, direction change	Poor generalization to unseen traffic scenes
CNN-based frame classification	Learns spatial accident patterns directly from frames	Ignores temporal context across frames
CNN-LSTM / 3D-CNN video models	Captures both spatial and temporal accident cues	Higher computational cost; needs larger labelled video data
Transfer learning (ResNet/VGG/MobileNet)	Leverages pretrained features to reduce training data needs	May need fine-tuning for traffic-specific scenes

- Research gap identified: most existing methods either capture spatial patterns (CNN-only) or handcrafted motion cues, but do not jointly model spatial and temporal accident dynamics efficiently at low computational cost. A CNN-LSTM hybrid was selected for this case study as it balances detection accuracy with feasibility for a project-scale implementation.

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8. Methodology / Working Process

- The methodology followed includes problem analysis, dataset collection, preprocessing (frame extraction, resizing, normalization, augmentation), model design (CNN-LSTM), training with the Adam optimizer and binary cross-entropy loss, evaluation against a baseline CNN, and discussion of deployment considerations.
- Block diagram / workflow:
Video/Image Input (CCTV) → Preprocessing (Frame Extraction, Resize, Normalize) → CNN Feature Extraction → Temporal Modeling (LSTM) → Accident Classification → Alert to Emergency Services

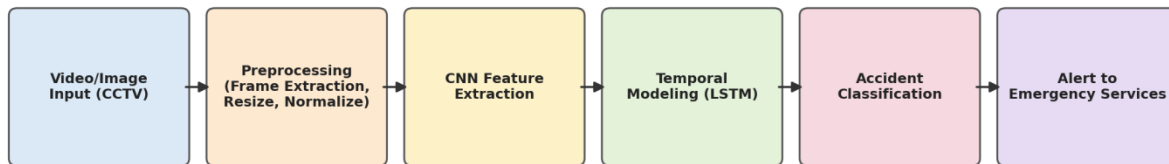


Fig. 1 — Proposed System Workflow for Real-Time Road Accident Detection

9. System Architecture / Framework

- Architecture diagram (CNN-LSTM model):
Input Frame (224×224×3) → Conv2D+ReLU (32 filters) → MaxPool → Conv2D+ReLU (64 filters) → MaxPool → Conv2D+ReLU (128 filters) → Global Average Pool → LSTM (temporal sequence) → Dense + Dropout → Sigmoid Output (Accident / No Accident)

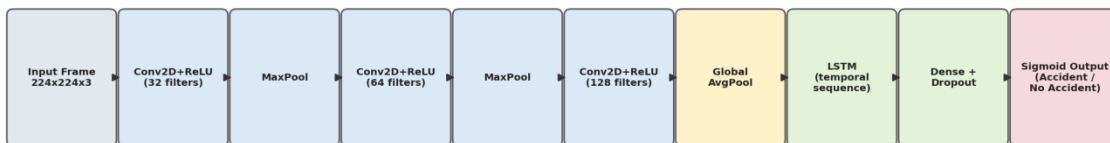


Fig. 2 — Proposed CNN-LSTM Model Architecture for Accident Detection

- Components: a CNN backbone (three convolution + max-pooling blocks) for per-frame spatial feature extraction, an LSTM layer to model temporal dependencies across a sequence of frames, and a dense layer with sigmoid activation for binary classification.
- Data flow: raw CCTV video → sampled frames → CNN feature vectors per frame → sequence of feature vectors fed to the LSTM → single accident/no-accident probability per sequence → thresholded decision → alert trigger to a monitoring dashboard / emergency services.

10. Algorithms / Techniques Used

- AI algorithms: supervised binary classification for event (accident) detection.
- Machine Learning models: baseline CNN classifier used for performance comparison.



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- Deep Learning architecture: CNN (spatial feature extraction) combined with LSTM (temporal sequence modeling) – a CNN-LSTM hybrid.
- Networking / deployment protocols: integration with CCTV/RTSP video streams and REST-based alert notifications to emergency-dispatch dashboards.

11. Tools, Software & Hardware Requirements

- Programming Language: Python 3.x
- IDE: Jupyter Notebook / Google Colab
- Frameworks: TensorFlow/Keras or PyTorch
- Libraries: OpenCV, NumPy, Pandas, Matplotlib, scikit-learn
- Hardware specifications: GPU-enabled system (NVIDIA GTX/RTX series or Google Colab GPU runtime), minimum 8 GB RAM, and a webcam/sample CCTV footage for real-time testing.

12. Dataset Used (if applicable)

- Dataset source: publicly available CCTV road-accident image/video datasets (e.g., accident-detection datasets on Kaggle).
- Number of samples: dataset size to be finalized during data collection; a reasonably balanced mix of accident and non-accident clips is targeted (illustrative planning split used in this study: ~38% accident frames, ~62% non-accident frames).
- Features: raw RGB video frames resized to 224×224, from which the CNN backbone automatically learns spatial features (edges, shapes, motion blur patterns).
- Train-Test split: frames/clips split into train, validation and test sets with attention to class balance between accident and non-accident samples (typical split: 70% train / 15% validation / 15% test).

13. Real-Time Example / Case Description

Consider a highway CCTV camera streaming live video to a traffic-control room. The proposed system continuously samples frames from the feed, extracts spatial features using the CNN backbone, and passes short sequences of these features to the LSTM layer. When the model detects a pattern consistent with a collision (e.g., a rear-end or side-impact collision, or a vehicle overturning), it raises a classification score above threshold, logs the event with a timestamp and camera ID, and sends an automated alert to the traffic-control dashboard for human verification before dispatching ambulance or traffic-police assistance.

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Illustrative Example of Input Frame Categories (schematic, not actual dataset samples)

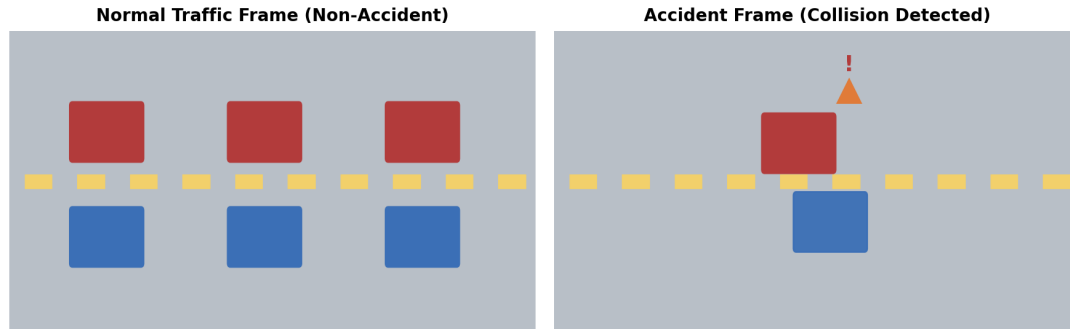


Fig. 3 — Illustrative Example of Input Frame Categories (schematic, not actual dataset samples)

14. Comparative Study

Existing Method	Proposed Method	Improvement
Motion/optical-flow detection	CNN-LSTM hybrid	Less sensitive to camera shake/lighting; learns patterns automatically
Classical ML on handcrafted features	CNN-LSTM hybrid	Better generalization to unseen traffic scenes
CNN-only frame classification	CNN-LSTM hybrid	Captures temporal context across frames, reducing false alarms

15. Applications

- Integration with highway/intersection CCTV networks for automated accident alerts
- Smart-city traffic-management dashboards for real-time incident monitoring
- Faster ambulance/traffic-police dispatch via automated notifications
- Extendable to toll plazas, tunnels, and school-zone crossings

16. Advantages

- Removes dependence on manual, fatigue-prone monitoring
- Operates continuously, 24/7, without attention lapses
- CNN+LSTM combo captures both spatial and temporal cues → more reliable than single-frame detection
- Works with existing CCTV infrastructure — no new hardware needed

17. Limitations

- Sensitive to lighting, weather, and camera-angle variability
- Class imbalance — accidents are rare vs. normal traffic footage
- Higher computational cost than a plain CNN due to temporal processing
- Model/dataset still at planning stage — not yet implemented or validated



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18. Future Scope

- Use lightweight backbones (e.g., MobileNet) for faster edge inference
- Add severity classification (minor vs. major collision)
- Multi-camera fusion to reduce blind spots/occlusion
- Direct integration with emergency-dispatch systems for automated routing

19. Challenges Faced

- Scarcity of large, diverse, labelled accident-video datasets
- Class imbalance making naive training unreliable (high accuracy, low recall)
- Trade-off between model accuracy and real-time inference speed
- Ensuring generalization across lighting, weather, and camera conditions

20. Expected Outcome

- Optimized CNN-LSTM model evaluated on Accuracy, Precision, Recall, F1-score, ROC-AUC
- Results interpreted in context (cost of false negatives vs. false positives)
- Direct comparison with baseline CNN to quantify benefit of temporal modeling
- Evidence-based justification for CNN-LSTM as a practical, deployable solution

21. SDG Mapping

- SDG Number: SDG 11
- SDG Name: Sustainable Cities and Communities
- Contribution: the system contributes to safer, more sustainable cities by improving urban road safety and emergency-response efficiency through automated, real-time accident detection.

22. Ethical Considerations (for AI-based topics)

- Privacy: raw surveillance footage must be access-restricted to authorized personnel, with retention limited to what is necessary for incident verification.
- Bias: the training dataset should be diverse across road types, vehicle types, lighting conditions and geographic regions to avoid biased detection performance.
- Security: video feeds and alert data must be protected against unauthorized access or tampering to prevent false alarms or missed detections.
- Transparency: the model's decisions should be interpretable enough to avoid unnecessary panic or misallocation of emergency resources due to false alarms, and any automated alert should ideally be verified by a human operator before large-scale emergency resources are deployed.



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23. Conclusion

This case study establishes the problem understanding, literature background and proposed design methodology for a Real-Time Road Accident Detection system using a CNN-LSTM based deep learning architecture. The proposed approach combines spatial and temporal feature learning to distinguish accident events from routine traffic footage, supporting faster emergency response and aligning with SDG 11 – Sustainable Cities and Communities. Future work will focus on full dataset acquisition, model implementation, training, quantitative evaluation against the baseline, and refinement based on observed results.

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